

Project Overview:

This project focuses on the analytical study of global meteorite landings using historical data spanning more than a century. The objective is to transform raw scientific data into an interactive and interpretable dashboard that highlights **temporal trends, geographical distribution, meteorite classification, and potential impact risk**.

Using Power BI, the project applies structured data modeling, classification logic, and visualization techniques to provide meaningful insights into meteorite occurrences worldwide. Special emphasis is placed on **explaining analytical assumptions**, such as risk categorization and meteorite classification, to ensure transparency and clarity in interpretation.

The project demonstrates the practical application of data analytics and visualization in understanding complex scientific datasets and communicating insights effectively to both technical and non-technical audiences.

Dataset Description:

Link(dataset)

<https://www.kaggle.com/datasets/nasa/meteorite-landings>

The dataset used in this project is the **Meteorite Landings dataset**, sourced from **NASA Open Data**, which provides verified records of meteorite events observed or discovered across the globe between **1700 and 2020**, with **45k row and 10 columns**.

The dataset contains scientifically documented meteorite records and includes key attributes such as:

- **Year of landing or discovery**
- **Name of the Meteorite**
- **Unique Meteorite id**
- **Meteorite mass (in grams)**
- **Geographical coordinates (latitude and longitude)**
- **Fall status** indicating whether the meteorite was observed during atmospheric entry (*Fell*) or discovered later on the ground (*Found*)

- **Meteorite type and composition**, including Stony, Iron, and Stony-Iron classifications

These attributes enable detailed analysis of meteorite behaviour across time, space, and physical characteristics. The presence of precise geographical coordinates allows continent-level mapping, while mass information supports risk classification and impact assessment.

Overall, the dataset provides a reliable and comprehensive foundation for analysing historical meteorite landings and deriving insights related to **distribution patterns, compositional differences, and impact risk**.

Objective of the Project:

1 - Analyse Global Distribution of Meteorite Landings

To study how meteorite landings are geographically distributed across the world using an interactive world map, identifying regions with higher meteorite occurrence.

2 - Understand Temporal Trends Using Decade-wise Analysis

To analyse how meteorite landings have changed over time (by decades), helping identify historical patterns and peak periods of meteorite activity.

3 - Classify Meteorites Based on Mass and Risk Level

To categorize meteorites into size-based and risk-based groups using mass data, enabling assessment of potential impact severity and hazard levels.

4 - Evaluate Meteorite Composition and Type Distribution

To examine different meteorite types (e.g., stony, iron, unclassified) and understand their frequency and dominance within the dataset.

5 - Identify High-Risk and Extreme Meteorite Events

To highlight meteorites with high and extreme risk factors based on mass, supporting impact analysis and scientific risk assessment.

6 - Enable Interactive Data Exploration Through Dynamic Filters

To allow users to interactively explore meteorite data using slicers (decade-based), ensuring KPIs, maps, and visuals update dynamically for deeper insights.

Data Preprocessing:

Step 1: Understanding the dataset

*Analyse row and columns carefully.

*Dataset columns - name, id, recclass, nametypes, fall, mass(g), year, reclang, reclat, Geolocation.

*Some of the columns like year, reclang, reclat and geolocation have missing values.

Column	Data Type
Name	Text
Year	Whole Number
Mass (g)	Decimal Number
reclat	Decimal Number
reclang	Decimal Number
recclass	Text
Id	Whole Number

Step – 2: Renaming the Columns.

Before: -

	A	B	C	D	E	F	G	H	I	J	K
1	name	id	nametype	recclass	mass (g)	fall	year	reclat	reclong	GeoLocation	
2	Aachen	1	Valid	L5	21	Fell	1880	50.775	6.08333	(50.775, 6.08333)	
3	Aarhus	2	Valid	H6	720	Fell	1951	56.18333	10.23333	(56.18333, 10.23333)	
4	Abee	6	Valid	EH4	107000	Fell	1952	54.21667	-113	(54.21667, -113.0)	
5	Acapulco	10	Valid	Acapulcoit	1914	Fell	1976	16.88333	-99.9	(16.88333, -99.9)	
6	Achiras	370	Valid	L6	780	Fell	1902	-33.16667	-64.95	(-33.16667, -64.95)	
7	Adhi Kot	379	Valid	EH4	4239	Fell	1919	32.1	71.8	(32.1, 71.8)	
8	Adzhi-Bog	390	Valid	LL3-6	910	Fell	1949	44.83333	95.16667	(44.83333, 95.16667)	
9	Agen	392	Valid	H5	30000	Fell	1814	44.21667	0.61667	(44.21667, 0.61667)	
10	Aguada	398	Valid	L6	1620	Fell	1930	-31.6	-65.23333	(-31.6, -65.23333)	
11	Aguila Blar	417	Valid	L	1440	Fell	1920	-30.86667	-64.55	(-30.86667, -64.55)	
12	Aiou el Af	423	Valid	Diogenite-	1000	Fell	1974	16.39806	-9.57028	(16.39806, -9.57028)	
13	AA'r	424	Valid	L6	24000	Fell	1925	19.08333	8.38333	(19.08333, 8.38333)	
14	Aire-sur-la	425	Valid	Unknown		Fell	1769	50.66667	2.33333	(50.66667, 2.33333)	
15	Akaba	426	Valid	L6	779	Fell	1949	29.51667	35.05	(29.51667, 35.05)	
16	Akbarpur	427	Valid	H4	1800	Fell	1838	29.71667	77.95	(29.71667, 77.95)	
17	Akwanga	432	Valid	H	3000	Fell	1959	8.91667	8.43333	(8.91667, 8.43333)	
18	Akyumak	433	Valid	Iron, IVA	50000	Fell	1981	39.91667	42.81667	(39.91667, 42.81667)	
19	Al Rais	446	Valid	CR2-an	160	Fell	1957	24.41667	39.51667	(24.41667, 39.51667)	
20	Al Zarnkh	447	Valid	LL5	700	Fell	2001	13.66033	28.96	(13.66033, 28.96)	
21	Alais	448	Valid	C11	6000	Fell	1806	44.11667	4.08333	(44.11667, 4.08333)	
22	Albareto	453	Valid	L/LL4	2000	Fell	1766	44.65	11.01667	(44.65, 11.01667)	
23	Alberta	454	Valid	L	625	Fell	1949		2	(2.0, 22.66667)	
24	Alby sur Cf	458	Valid	Eucrite-mr	252	Fell	2002	45.82133	6.01533	(45.82133, 6.01533)	
25	Alsdrworth	461	Valid	LL5	700	Fell	1835	51.78333	-1.78333	(51.78333, -1.78333)	
26	Aleppo	462	Valid	L6	3200	Fell	1873	36.23333	37.13333	(36.23333, 37.13333)	
27	Alessandri	463	Valid	H5	908	Fell	1860	44.88333	8.75	(44.88333, 8.75)	
28	Alexandro	465	Valid	H4	9251	Fell	1900	50.95	31.81667	(50.95, 31.81667)	
29	Alfianello	466	Valid	L6	228000	Fell	1883	45.26667	10.15	(45.26667, 10.15)	
30	Allegan	2276	Valid	H5	32000	Fell	1899	42.53333	-85.88333	(42.53333, -85.88333)	
31	Allende	2278	Valid	CV3	2000000	Fell	1969	26.96667	-105.31667	(26.96667, -105.31667)	
32	Almahata	48915	Valid	Ureilite-an	3950	Fell	2008	20.74575	32.41275	(20.74575, 32.41275)	
33	Almahata	3984	Valid	H6	6000	Fell	1973	35.33333	44.33333	(35.33333, 44.33333)	
<	>	Meteorite Landings		+							

After Renaming:

File Home Help **Table tools**

Name
 Manage relationships
 New measure
 Quick measure
 New column
 New table
 Mark a

Meteorite Name	Meteorite id	Meteorite Type	Meteorite Class	Mass (grams)	Fall status	Year	Latitude(*)	Longitude(*)	GeoLocation
Yamato 981200	37822	Valid	H4	12.23	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981115	37737	Valid	H4	12.59	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981116	37738	Valid	H4	13.47	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981120	37742	Valid	H4	14.79	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981123	37745	Valid	H4	6.26	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981150	37772	Valid	H4	7.75	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981159	37781	Valid	H4	8.85	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981167	37789	Valid	H4	31.29	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981169	37791	Valid	H4	8.34	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981173	37795	Valid	H4	8.92	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981219	37841	Valid	H4	23.65	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981222	37844	Valid	H4	12.62	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981223	37845	Valid	H4	25.82	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981224	37846	Valid	H4	27.15	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981229	37851	Valid	H4	46.09	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981245	37867	Valid	H4	29.08	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981252	37874	Valid	H4	11.51	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981299	37921	Valid	H4	31.16	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981307	37929	Valid	H4	68.91	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981309	37931	Valid	H4	5.34	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981319	37941	Valid	H4	7.73	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981321	37943	Valid	H4	12.73	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981333	37955	Valid	H4	133.65	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981361	37983	Valid	H4	13.42	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981381	38003	Valid	H4	5.59	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981418	38040	Valid	H4	6.67	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981425	38047	Valid	H4	15.19	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981436	38058	Valid	H4	26.42	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981437	38059	Valid	H4	5.15	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981438	38060	Valid	H4	370.88	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981453	38075	Valid	H4	8.76	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981455	38077	Valid	H4	7.25	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981456	38078	Valid	H4	11.58	Found	1998	0	0	0 (0.0, 0.0)
Yamato 981457	38079	Valid	H4	17.46	Found	1998	0	0	0 (0.0, 0.0)

Table: Meteorite_Landings (37,500 rows)

Step-3:

Handling missing value:

*Some of the columns like Year , Longitude, Latitude have missing values do I delete those column that have missing values and clean the dataset.

**Applied Step:

The screenshot displays the Microsoft Power BI Desktop interface. The main view shows a table named 'Meteorite_Landings' with the following columns: Name, Mass (grams), Fall_status, Year, Latitude, Longitude, and GeoLocation. The table contains 23 rows of data. The right-hand pane shows the 'Query Settings' for 'Meteorite_Landings', including 'Properties' and 'Applied Steps'. The 'Applied Steps' list includes: Source, Promoted Headers, Changed Type, Renamed Columns, Removed Blank Rows, Filtered Rows, Filtered Rows1, Filtered Rows2, Filtered Rows3, and Filtered Rows4. The bottom status bar indicates '10 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

Step - 4: Adding new column using DAX function.

1-Decade:

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Help

Table tools

Column tools

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Type	Meteorite Class	Mass (grams)	Fall_status	Year	Latitude(°)	Longitude(°)	GeoLocation	Decade	Mass Category	Continent	Meteorite Type Group	Risk Factor
H4	12.23	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	12.59	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	13.47	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	14.79	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	6.26	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	7.75	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	8.85	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	31.29	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	8.34	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	8.92	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	23.65	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	12.62	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	25.82	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	27.15	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	46.09	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	29.08	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	11.51	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	31.16	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	68.91	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	5.34	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	7.73	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	12.73	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	133.65	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	13.42	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	5.59	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	6.67	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	15.19	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	26.42	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	5.15	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	370.88	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	8.76	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	7.25	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	

Table: Meteorite_Landings (37,500 rows) Column: Decade (13 distinct values)

Sign in

Share

Data

Search

Meteorite_Landings

Average Mass (g)

Continent

Decade

Fall_status

GeoLocation

High Risk Meteorites

Latitude(°)

Longitude(°)

Mass (grams)

Mass Category

Max Mass (g)

Meteorite id

Meteorite Name

Meteorite Type

Meteorite Type Group

Meteorite Class

Risk Factor

Total Meteorites

Verified Meteorites

Year

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Table

Chart

Map

1Continent =
2SWITCH(
3TRUE(),
4
5'Meteorite_Landings'[Latitude(°)] >= 15 &&
6'Meteorite_Landings'[Latitude(°)] <= 75 &&
7'Meteorite_Landings'[Longitude(°)] >= -170 &&
8'Meteorite_Landings'[Longitude(°)] <= -50,
9"North America",
10
11'Meteorite_Landings'[Latitude(°)] >= -60 &&
12'Meteorite_Landings'[Latitude(°)] <= 15 &&
13'Meteorite_Landings'[Longitude(°)] >= -90 &&
14'Meteorite_Landings'[Longitude(°)] <= -30,
15"South America",
16
17'Meteorite_Landings'[Latitude(°)] >= 35 &&
18'Meteorite_Landings'[Latitude(°)] <= 75 &&
19'Meteorite_Landings'[Longitude(°)] >= -25 &&
20'Meteorite_Landings'[Longitude(°)] <= 45,
21"Europe",
22
23'Meteorite_Landings'[Latitude(°)] >= -35 &&
24'Meteorite_Landings'[Latitude(°)] <= 35 &&
25'Meteorite_Landings'[Longitude(°)] >= -20 &&
26'Meteorite_Landings'[Longitude(°)] <= 55,
27"Africa",
28
29'Meteorite_Landings'[Latitude(°)] >= 5 &&
30'Meteorite_Landings'[Latitude(°)] <= 80 &&
31'Meteorite_Landings'[Longitude(°)] >= 45 &&
32'Meteorite_Landings'[Longitude(°)] <= 180,
33"Asia",
34
35'Meteorite_Landings'[Latitude(°)] >= -50 &&
36'Meteorite_Landings'[Latitude(°)] <= -5 &&
37'Meteorite_Landings'[Longitude(°)] >= 110 &&
38'Meteorite_Landings'[Longitude(°)] <= 180,
39

Table: Meteorite_Landings (37,500 rows) Column: Continent (8 distinct values)

Type	Meteorite Class	Mass (grams)	Fall_status	Year	Latitude(°)	Longitude(°)	GeoLocation	Decade	Mass Category	Continent	Meteorite Type Group	Risk Factor
H4		12.23	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.59	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk

Data

Search

Meteorite_Landings

Average Mass (g)

Continent

Decade

Fall_status

GeoLocation

High Risk Meteorites

Latitude(°)

Longitude(°)

Mass (grams)

Mass Category

Max Mass (g)

Meteorite id

Meteorite Name

Meteorite Type

Meteorite Type Group

Meteorite Class

Risk Factor

Total Meteorites

Verified Meteorites

Year

Table

ENG

IN

22:52

15-12-2023

3-Mass Category

dashboard(powerbi) • Last saved: Today at 8:27 pm

File Home Help Table tools Column tools

Name Risk Factor Data type Text Text % 0.00 → 0.0 Auto Σ Don't summarize Uncategorized

1 Risk Factor =
2 SWITCH(
3 TRUE(),
4 'Meteorite_Landings'[Mass (grams)] < 1000, "Low Risk",
5 'Meteorite_Landings'[Mass (grams)] < 10000, "Moderate Risk",
6 'Meteorite_Landings'[Mass (grams)] < 100000, "High Risk",
7 "Extreme Risk"
8)
9

Type	Meteorite Class	Mass (grams)	Fall_status	Year	Latitude(°)	Longitude(°)	GeoLocation	Decade	Mass Category	Continent	Meteorite Type Group	Risk Factor
H4		12.23	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.59	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		13.47	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		14.79	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		6.26	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		7.75	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.85	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		31.29	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.34	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.92	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		23.65	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.62	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		25.82	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		27.15	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		46.09	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		29.08	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		11.51	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		31.16	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		68.91	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		5.34	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		7.73	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.73	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		133.65	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		13.42	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		5.59	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		6.67	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		15.19	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk

Table: Meteorite_Landings (37,500 rows) Column: Risk Factor (4 distinct values)

12°C Clear Search ENG IN 22:52 15-12-2025

4- Meteorite Type group

dashboard(powerbi) • Last saved: Today at 8:27 pm

File Home Help Table tools Column tools

Name Meteorite Type ... Data type Text Text % 0.00 → 0.0 Auto Σ Don't summarize Uncategorized

1 Meteorite Type Group =
2 SWITCH(
3 TRUE(),
4
5 CONTAINSSTRING('Meteorite_Landings'[Meteorite Class], "Iron"), "Iron",
6
7 CONTAINSSTRING('Meteorite_Landings'[Meteorite Class], "Pallasite") ||
8 CONTAINSSTRING('Meteorite_Landings'[Meteorite Class], "Mesosiderite"),
9 "Stony-Iron",
10
11 LEFT('Meteorite_Landings'[Meteorite Class], 2) = "LL", "Stony (Chondrite - LL)",
12 LEFT('Meteorite_Landings'[Meteorite Class], 1) = "H", "Stony (Chondrite - H)",
13 LEFT('Meteorite_Landings'[Meteorite Class], 1) = "L", "Stony (Chondrite - L)",
14
15 CONTAINSSTRING('Meteorite_Landings'[Meteorite Class], "Achondrite"), "Stony (Achondrite)",
16 CONTAINSSTRING('Meteorite_Landings'[Meteorite Class], "Stone"), "Stony",
17 "Other"
18)
19

Type	Meteorite Class	Mass (grams)	Fall_status	Year	Latitude(°)	Longitude(°)	GeoLocation	Decade	Mass Category	Continent	Meteorite Type Group	Risk Factor
H4		12.23	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.59	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		13.47	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		14.79	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		6.26	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		7.75	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.85	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		31.29	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.34	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.92	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		23.65	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.62	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		25.82	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		27.15	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		46.09	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		29.08	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		11.51	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		31.16	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		68.91	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk

Table: Meteorite_Landings (37,500 rows) Column: Meteorite Type Group (8 distinct values)

12°C Clear Search ENG IN 22:52 15-12-2025

5-Risk Factor

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Sign in

FileHomeHelpTable toolsColumn tools

NameRisk FactorData typeTextText\$%‰AutoΣDon't summarizeUncategorized

1 Risk Factor =
2 SWITCH(
3 TRUE(),
4 'Meteorite_Landings'[Mass (grams)] < 1000, "Low Risk",
5 'Meteorite_Landings'[Mass (grams)] < 10000, "Moderate Risk",
6 'Meteorite_Landings'[Mass (grams)] < 100000, "High Risk",
7 "Extreme Risk"
8)
9

Type	Meteorite Class	Mass (grams)	Fall status	Year	Latitude(°)	Longitude(°)	GeoLocation	Decade	Mass Category	Continent	Meteorite Type Group	Risk Factor
H4		12.23	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.59	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		13.47	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		14.79	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		6.26	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		7.75	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.85	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		31.29	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.34	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.92	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		23.65	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.62	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		25.82	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		27.15	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		46.09	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		29.08	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		11.51	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		31.16	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		68.91	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		5.34	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		7.73	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.73	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		133.65	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		13.42	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		5.59	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		6.67	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		15.19	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk

Table: Meteorite_Landings (37,500 rows) Column: Risk Factor (4 distinct values)

12°C
Clear

Search

ENG
IN

22:52
15-12-2025

Search

Meteorite_Landings

- Average Mass (g)
- Continent
- Decade
- Fall_status
- GeoLocation
- High Risk Meteorites
- Latitude(°)
- Longitude(°)
- Mass (grams)
- Max Mass (g)
- Mass Category
- Meteorite id
- Meteorite Name
- Meteorite Type
- Meteorite Type Group
- Meteorite Class

Risk Factor

- Total Meteorites
- Verified Meteorites
- Year

Table

Step-6

Finding some Measure for Dashboard

1- Verified Meteorites

dashboard(powerbi) • Last saved: Today at 8:27 pm

File Home Help Table tools Measure tools

Name Verified Meteorite... Home table Meteorite_Land... Format Whole number \$ % 0 Data category Uncategorized

1 Verified Meteorites =
2 CALCULATE(
3 COUNTROWS('Meteorite_Landings'),
4 "Meteorite_Landings'[Fall_status]' = "Fall"
5)
6

Type	Meteorite Class	Mass (grams)	Fall_status	Year	Latitude(°)	Longitude(°)	GeoLocation	Decade	Mass Category	Continent	Meteorite Type Group	Risk Factor
H4		12.23	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.59	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		13.47	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		14.79	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		6.26	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		7.75	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.85	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		31.29	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.34	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.92	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		23.65	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.62	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		25.82	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		27.15	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		46.09	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		29.08	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		11.51	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		31.16	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		68.91	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		5.34	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		7.73	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.73	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		133.65	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		13.42	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		5.59	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		6.67	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		15.19	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		26.42	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		5.15	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		370.88	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk

Table: Meteorite_Landings (37,500 rows) Column: Verified Meteorites (0 distinct values)

2-Max Mass

dashboard(powerbi) • Last saved: Today at 8:27 pm

File Home Help Table tools Measure tools

Name Max Mass (g) Home table Meteorite_Land... Format General \$ % 0 Auto Data category Uncategorized

1 Max Mass (g) =
2 MAX('Meteorite_Landings'[Mass (grams)])
3

Type	Meteorite Class	Mass (grams)	Fall_status	Year	Latitude(°)	Longitude(°)	GeoLocation	Decade	Mass Category	Continent	Meteorite Type Group	Risk Factor
H4		12.23	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.59	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		13.47	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		14.79	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		6.26	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		7.75	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.85	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		31.29	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.34	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.92	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		23.65	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.62	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		25.82	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		27.15	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		46.09	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		29.08	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		11.51	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		31.16	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		68.91	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		5.34	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		7.73	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.73	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		133.65	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		13.42	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		5.59	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		6.67	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		15.19	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		26.42	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		5.15	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		370.88	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.76	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		7.25	Found	1998	0	0	0 (0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk

Table: Meteorite_Landings (37,500 rows) Column: Max Mass (g) (0 distinct values)

3-High Risk Meteorite

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FileHomeHelpTable toolsMeasure tools

NameHigh Risk Mete...Home tableMeteorite_Land...FormatWhole number\$ %0Data categoryUncategorized

1High Risk Meteorites =2CALCULATE(3COUNTROWS("Meteorite_Landings"),4'Meteorite_Landings'[Risk Factor] IN {"High Risk", "Extreme"}5)6

Type	Meteorite Class	Mass (grams)	Fall_status	Year	Latitude(°)	Longitude(°)	GeoLocation	Decade	Mass Category	Continent	Meteorite Type Group	Risk Factor
H4	12.23	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	12.59	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	13.47	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	14.79	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	6.26	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	7.75	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	8.85	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	31.29	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	8.34	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	8.92	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	23.65	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	12.62	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	25.82	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	27.15	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	46.09	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	29.08	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	11.51	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	31.16	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	68.91	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	5.34	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	7.73	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	12.73	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	133.65	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	13.42	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	5.59	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	6.67	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	15.19	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	26.42	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	5.15	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	
H4	370.88	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk	

Data

Search

Meteorite_Landings

Average Mass (g)

Continent

Decade

Fall_status

GeoLocation

High Risk Meteorites

Latitude(°)

Longitude(°)

Mass (grams)

Mass Category

Max Mass (g)

Meteorite id

Meteorite Name

Meteorite Type

Meteorite Type Group

Meteorite Class

Risk Factor

Total Meteorites

Verified Meteorites

Year

Table

Table: Meteorite_Landings (37,500 rows) Column: High Risk Meteorites (0 distinct values)

12°C ClearSearchENG IN22:5915-12-2025

4-Average Mass(g)

dashboard(powerbi) • Last saved: Today at 8:27 pm

FileHomeHelpTable toolsMeasure tools

NameAverage Mass (g)

Home tableMeteorite_Land...

FormatGeneral

\$ %

Auto

Data categoryUncategorized

1Average Mass (g) =

2AVERAGE('Meteorite_Landings'[Mass (grams)])

3

Type	Meteorite Class	Mass (grams)	Fall_status	Year	Latitude(°)	Longitude(°)	GeoLocation	Decade	Mass Category	Continent	Meteorite Type Group	Risk Factor
H4		12.23	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.59	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		13.47	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		14.79	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		6.26	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		7.75	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.85	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		31.29	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.34	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.92	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		23.65	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.62	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		25.82	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		27.15	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		46.09	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		29.08	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		11.51	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		31.16	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		68.91	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		5.34	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		7.73	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		12.73	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		133.65	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		13.42	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		5.59	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		6.67	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		15.19	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		26.42	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		5.15	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		370.88	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		8.76	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk
H4		7.25	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa	Stony (Chondrite - H)	Low Risk

Table: Meteorite_Landings (37,500 rows) Column: Average Mass (g) (0 distinct values)

Data

Search

Meteorite_Landings

Average Mass (g)

Continent

Decade

Fall_status

GeoLocation

High Risk Meteorites

Latitude(°)

Longitude(°)

Mass (grams)

Mass Category

Max Mass (g)

Meteorite id

Meteorite Name

Meteorite Type

Meteorite Type Group

Meteorite Class

Risk Factor

Total Meteorites

Verified Meteorites

Year

Table

5- Total Meteorites

dashbaord(powerbi) • Last saved: Today at 8:27 pm

File Home Help Table tools Measure tools

Name Total Meteorites Home table Meteorite_Land... Format Whole number \$ % 0 Data category Uncategorized

1 Total Meteorites =
2 COUNTROWS('Meteorite_Landings')
3

Meteorite Name	Meteorite id	Meteorite Type	Meteorite Class	Mass (grams)	Fall_status	Year	Latitude(°)	Longitude(°)	GeoLocation	Decade	Mass Category	Continent
Yamato 981200	37822	Valid	H4	12.23	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981115	37737	Valid	H4	12.59	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981116	37738	Valid	H4	13.47	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981120	37742	Valid	H4	14.79	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981123	37745	Valid	H4	6.26	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981150	37772	Valid	H4	7.75	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981159	37781	Valid	H4	8.85	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981167	37789	Valid	H4	31.29	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981169	37791	Valid	H4	8.34	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981173	37795	Valid	H4	8.92	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981219	37841	Valid	H4	23.65	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981222	37844	Valid	H4	12.62	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981223	37845	Valid	H4	25.82	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981224	37846	Valid	H4	27.15	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981229	37851	Valid	H4	46.09	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981245	37867	Valid	H4	29.08	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981252	37874	Valid	H4	11.51	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981299	37921	Valid	H4	31.16	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981307	37929	Valid	H4	68.91	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981309	37931	Valid	H4	5.34	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981319	37941	Valid	H4	7.73	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981321	37943	Valid	H4	12.73	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981333	37955	Valid	H4	133.65	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981361	37963	Valid	H4	13.42	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981381	38003	Valid	H4	5.59	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981418	38040	Valid	H4	6.67	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981425	38047	Valid	H4	15.19	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981436	38058	Valid	H4	26.42	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981437	38059	Valid	H4	5.15	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981438	38060	Valid	H4	370.88	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981453	38075	Valid	H4	8.76	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa
Yamato 981455	38077	Valid	H4	7.25	Found	1998	0	0	(0.0, 0.0)	1990s	Small	Africa

Table: Meteorite_Landings (37,500 rows) Column: Total Meteorites (0 distinct values)

Gold +0.63%

Search

ENG IN 23:06 15-12-2025

Step 7-

Adding a table that have meteorite type and compositions

dashbaord(powerbi) • Last saved: Today at 8:27 pm

File Home Help Table tools

Name Table Manage relationships New measure Quick measure New column New table Mark as date table

Meteorite Type Composition

Stony	Composition of Silicates(rocks)
Iron	Iron + Nickel metal
Stony-Iron	Rock +Iron(rare type)

Table: Table (3 rows)

Table: Meteorite_Landings (37,500 rows) Column: Total Meteorites (0 distinct values)

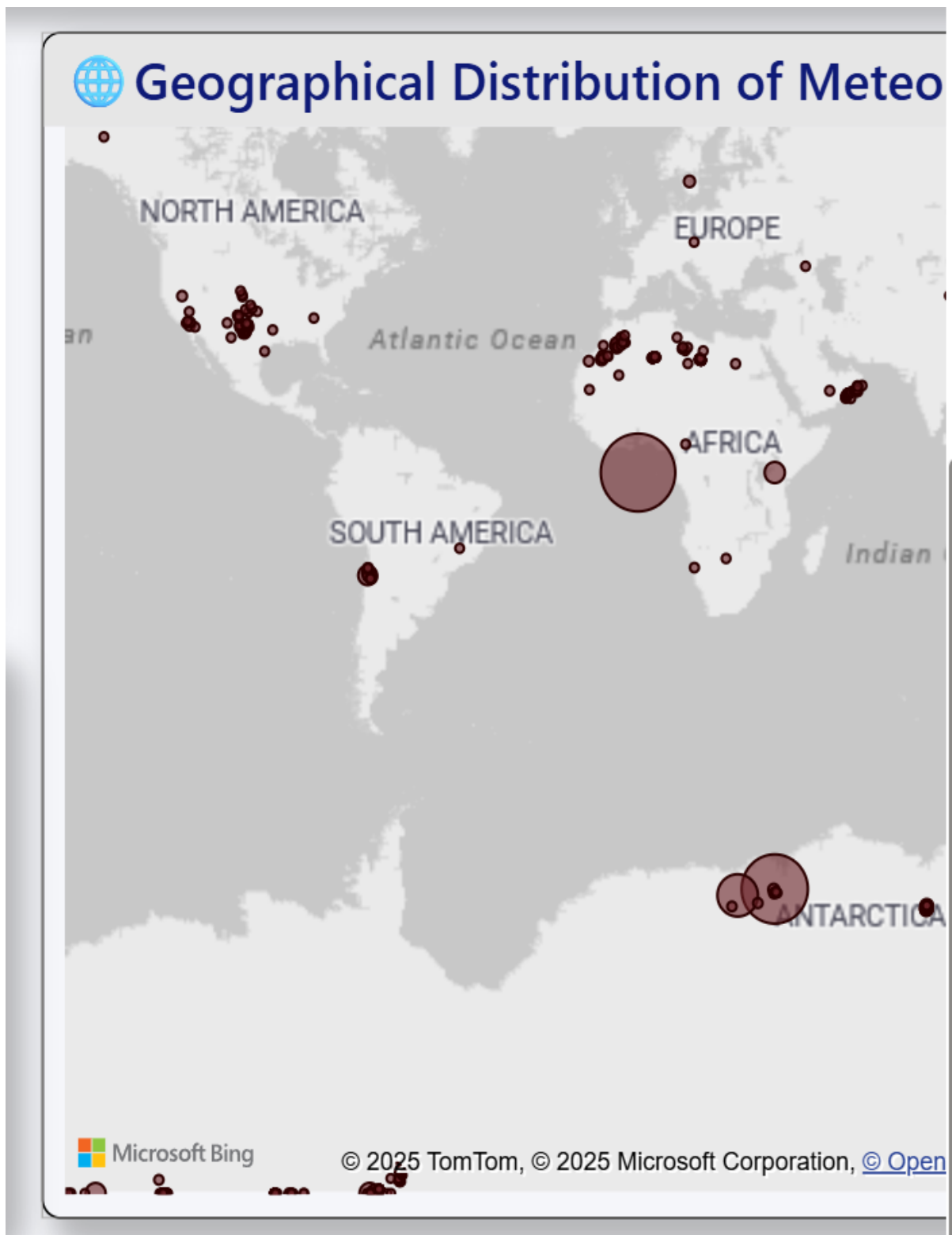
Gold +0.63%

Search

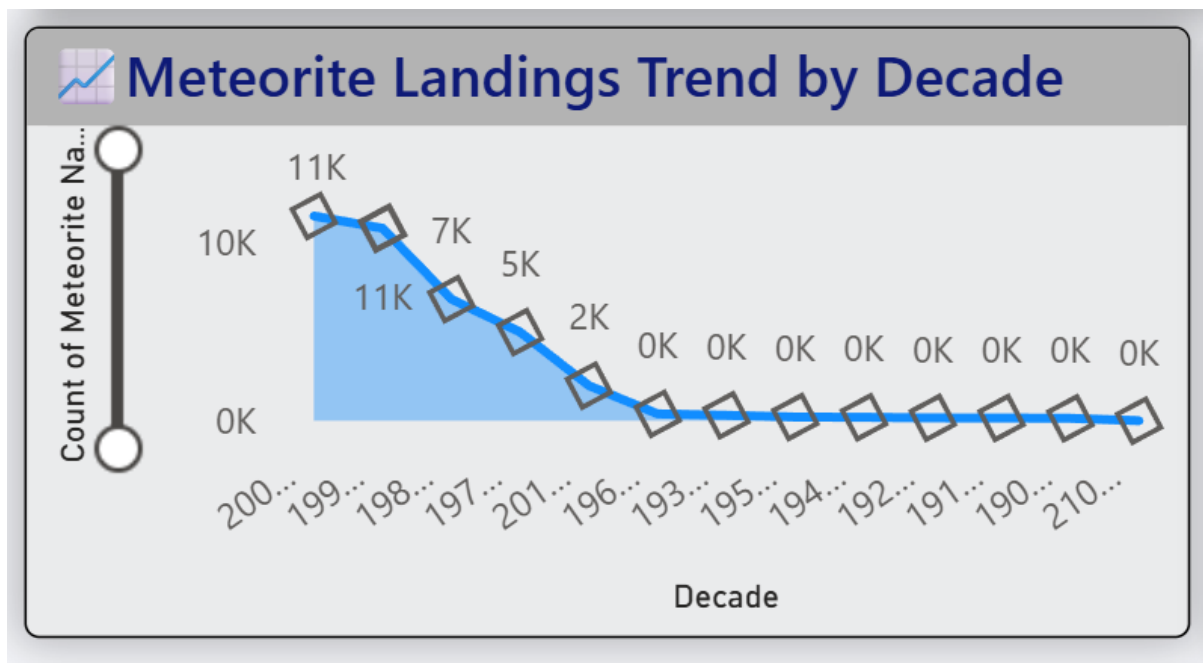
ENG IN 23:00 15-12-2025

Preparing Visuals

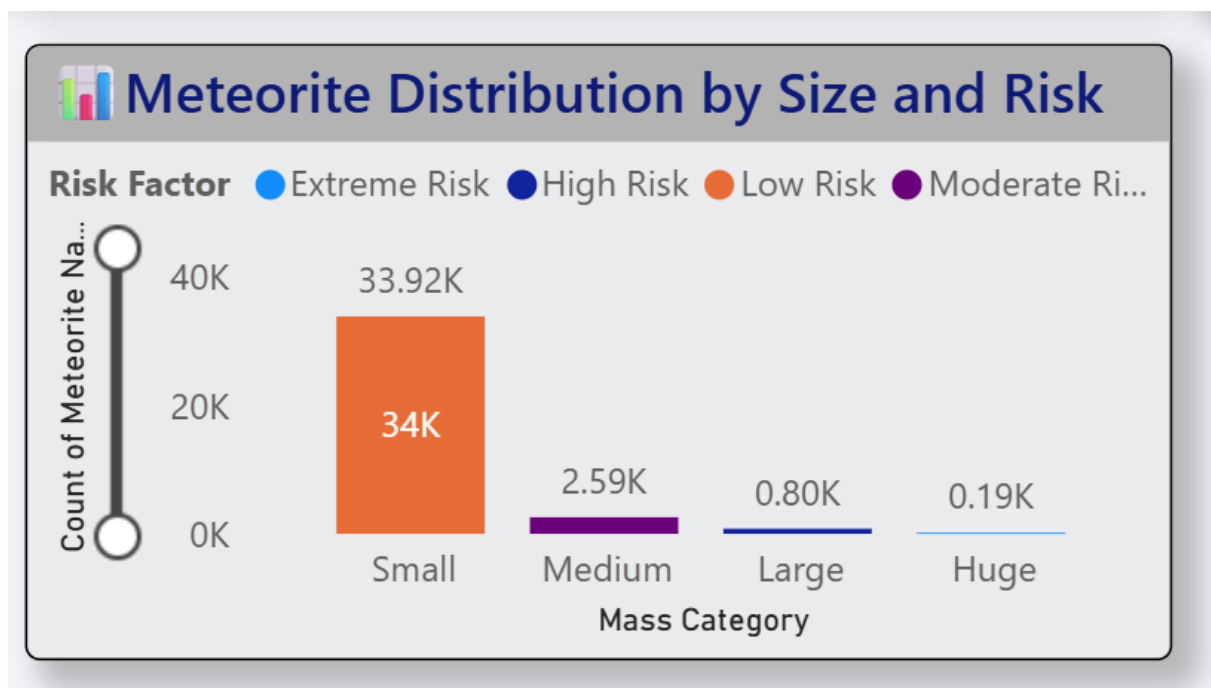
Objective 1- Analyse Global Distribution of Meteorite Landings



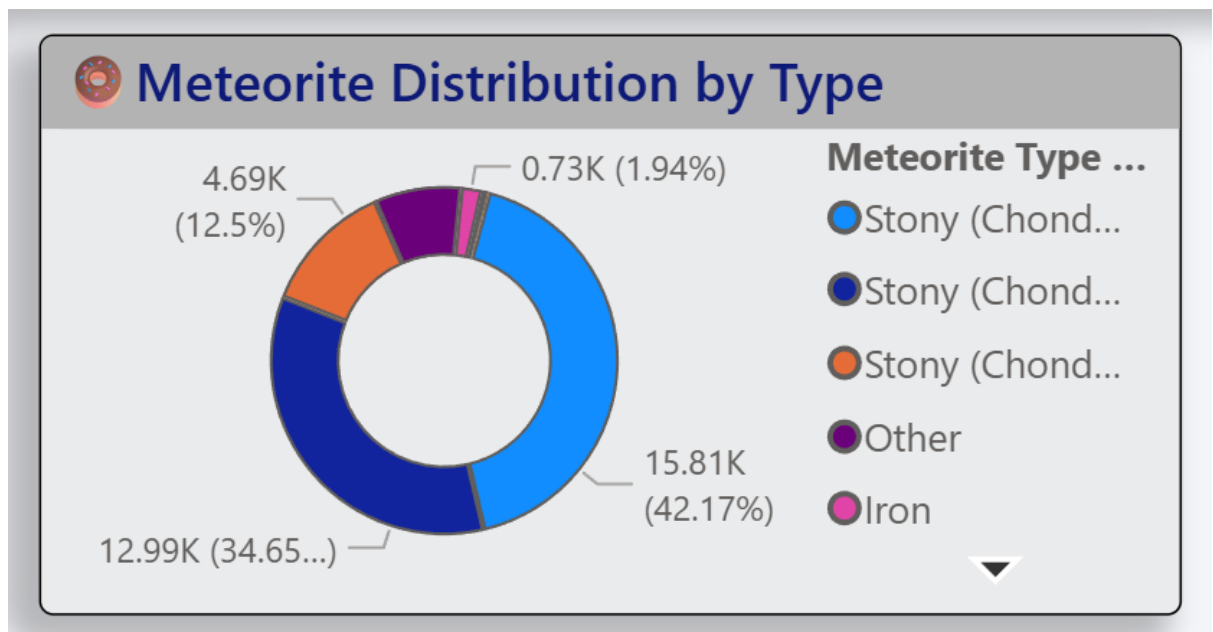
2- Understand Temporal Trends Using Decade-wise Analysis



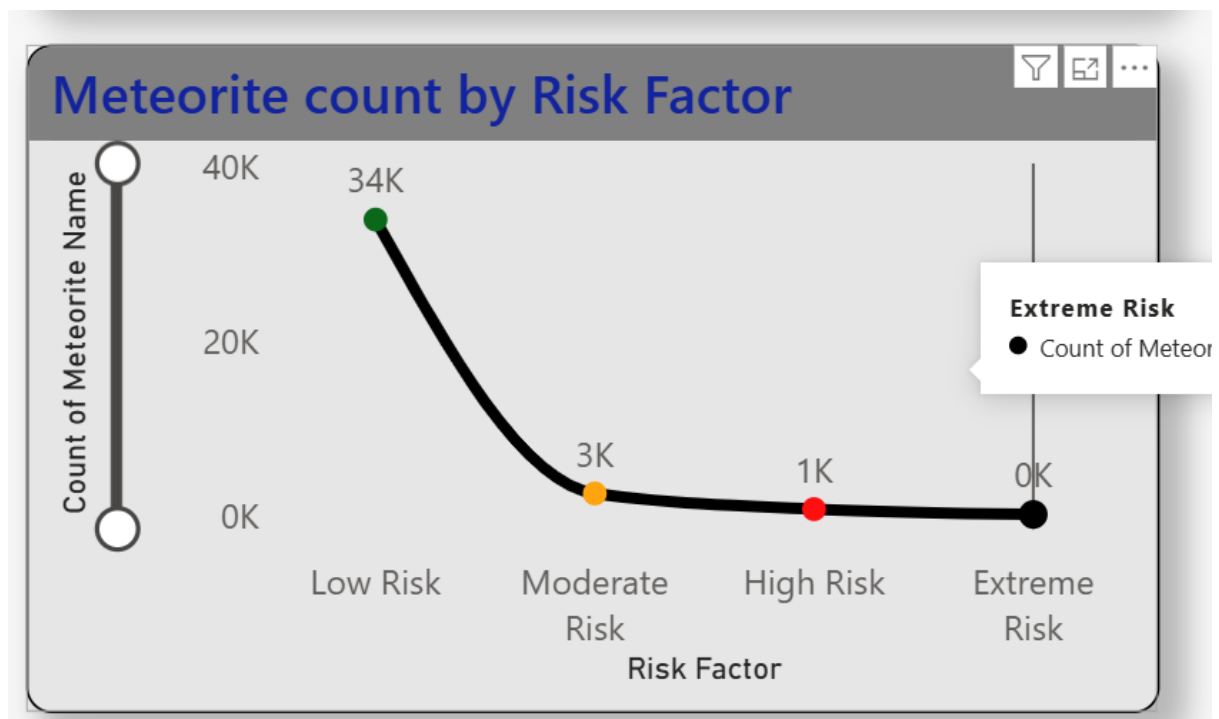
3-Classify Meteorites Based on Mass and Risk Level



4- Evaluate Meteorite Composition and Type Distribution

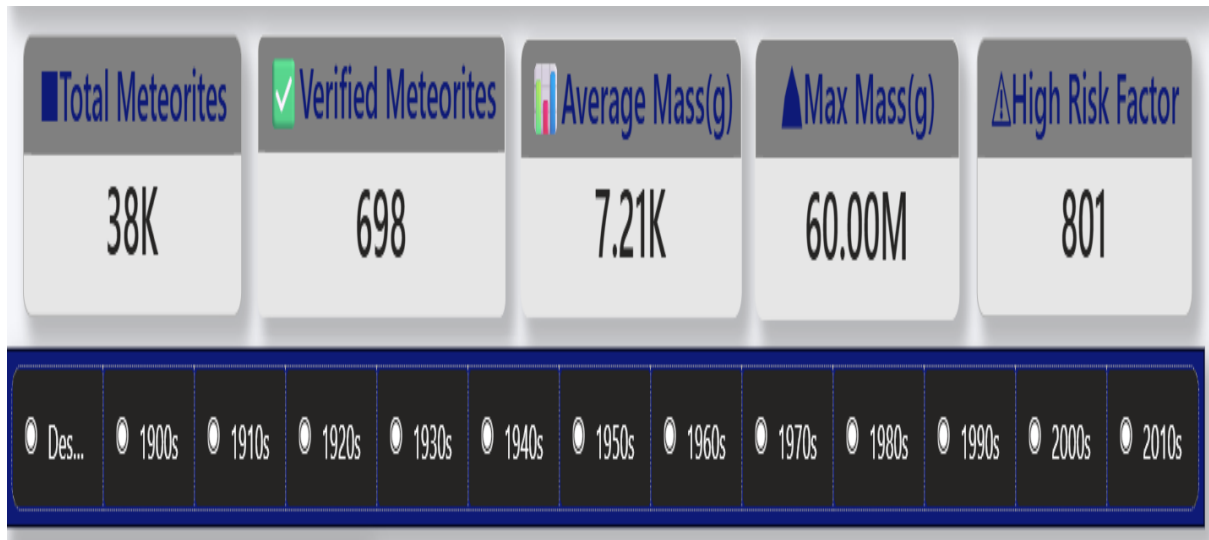


5- Identify High-Risk and Extreme Meteorite Events



6- Enable Interactive Data Exploration Through Dynamic Filters

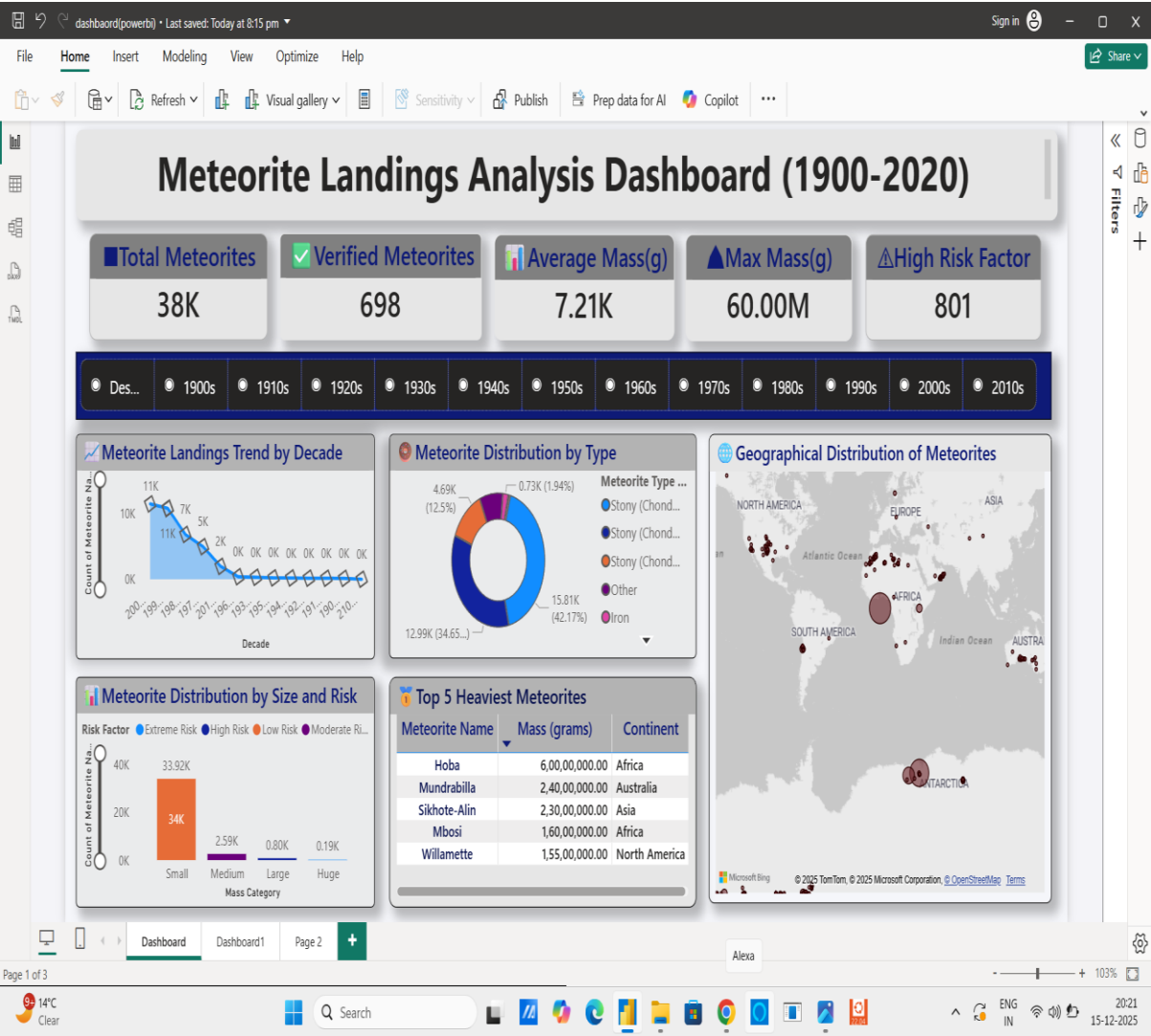
(using KPI and Slicer)

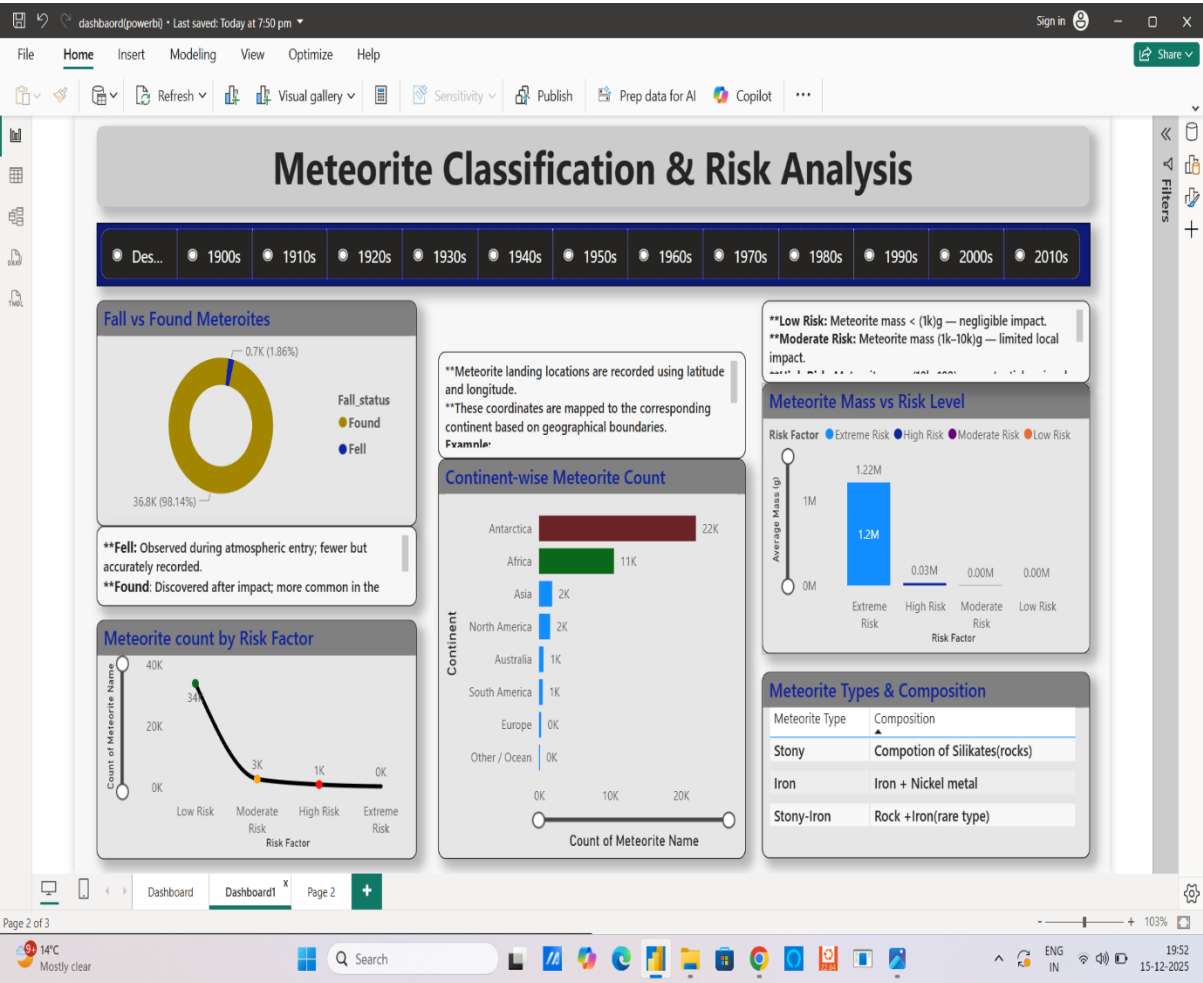


Final Dashboard:

Aligning, spacing and polishing all the visuals and make a interactive dashboard.

Page 1-





Insights:

Insight 1:

Meteorite landings are not uniformly distributed across the globe. Certain regions and continents show higher concentrations, indicating geographic and observational biases.

Insight 2:

Meteorite landings show significant variation across decades, with a noticeable increase in recent decades.

This increase is likely due to better detection, reporting systems, and scientific monitoring, rather than an actual rise in meteorite activity.

Insight 3:

Most meteorites fall into the Small and Medium mass categories, while Large and Extreme meteorites are rare.

However, even a small number of extreme-mass meteorites contribute disproportionately to overall risk.

Insight 4:

Stony meteorites dominate the dataset, making them the most commonly observed meteorite type.

Iron and Stony-Iron meteorites are comparatively rare but often have higher mass, increasing their potential risk

Insight 5:

High-risk meteorites are strongly correlated with mass, confirming that mass is a key indicator of impact severity.

Insight 6:

The top 5 meteorites by mass are outliers, significantly heavier than the average meteorite.

These extreme events are crucial for risk assessment and planetary defence research.

Summary:

This project presents an interactive Power BI dashboard analyzing global meteorite landing data.

The dashboard explores meteorite distribution across geography, time, mass, and risk categories.

Advanced preprocessing and DAX-based calculated columns were used to enrich the dataset and enable deeper insights.

The analysis highlights patterns in meteorite landings, identifies high-risk meteorites, and demonstrates how data visualization can support scientific and risk-based understanding of extraterrestrial events.

Overall, the project delivers a clean, interactive, and industry-style analytical dashboard suitable for academic and analytical use.